

Announcement 2: ExpansionRx - OpenADMET Blind Challenge

OpenADMET

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OPENADMET

Next generation ADMET modelling
for therapeutic development

Additional details on the coming challenge

Absorption, Distribution, Metabolism, Excretion and Toxicity–ADMET–properties can make or break preclinical and clinical development of small molecules. At [OpenADMET](#) we address the unpredictable nature of these properties through open science, generating experimental data and building predictive models of ADMET properties. A key component of these efforts is running community **blind challenges** to benchmark the current state of the art in ADMET modeling.

On September 16, we announced the [ExpansionRx-OpenADMET blind challenge](#) in partnership with Expansion Therapeutics and HuggingFace. Participants will predict multiple ADMET properties for a diverse set of molecules collected from a real-world drug discovery campaign targeting RNA-mediated diseases, including Myotonic Dystrophy, ALS, and Dementia. Below, we provide more details on the coming challenge, how to participate, and the dataset and evaluation criteria we will use to select the winners.

The Challenge

Participants will be tasked with solving real-world ADMET prediction problems ExpansionRx faced during lead optimization. Specifically, you will be asked to **predict the ADMET properties of late-stage molecules based on earlier-stage data** from the same campaigns.

This type of **time-split** challenge mimics the real-world task of leveraging historical optimization data to improve decision-making, a core component of multi-parameter optimization in drug development and a key point at which models can add value. Not an easy task!

You will be provided with molecules from early in the optimization campaign to train your model (**training split** is released on HuggingFace) and provided with molecules from later in the lead-opt campaign to then submit your predictions on (**test split**).

You can find a small sample of the training data here:

<https://huggingface.co/datasets/openadmet/openadmet-expansionrx-challenge-teaser>

Check out the HuggingFace Space where the challenge will be hosted:

<https://huggingface.co/spaces/openadmet/OpenADMET-ExpansionRx-Challenge>

Come and discuss the challenge on the [OpenADMET Discord](#).

OpenADMET will also be submitting a series of baselines to support participants, which will be shared using our open-source infrastructure. Stay tuned for updates on the Discord Channel!

Endpoints included in this challenge

You will apply your models for the prediction of nine (9) crucial ADMET endpoints:

- **LogD:** Measures compound's lipophilicity at a specific pH. Drugs typically fall into a well defined LogD range that balances aqueous solubility with membrane permeability, making understanding changes in LogD across a chemical series important to medicinal chemists. Additionally, assessing LogD of candidate molecules can suggest whether candidate molecules are "efficient" for their lipophilicity (lipophilicity generally has a linear relationship with affinity).
- **Kinetic Solubility (KSOL):** Measures how much a compound can be dissolved under non-equilibrium conditions. Helps screen compounds that will fail due to poor absorption or low bioavailability.
- **Human Liver Microsomal (HLM) stability:** Supports understanding of a compound's susceptibility to liver metabolism and can be used to predict *in-vivo* clearance of a candidate molecule. Measured using human liver microsomes and reported as intrinsic clearance for the compound Cl_{int} (mL/min/kg).
- **Mouse Liver Microsomal (MLM) stability:** Also used to predict *in-vivo* clearance of a candidate. The study of both MLM and RLM can provide a more comprehensive understanding of a compound's metabolic profile and how a compound might behave in multi-species preclinical development.
- **Caco-2 Papp A>B:** Measures the rate of flux of a compound across polarized Caco-2 cell monolayers from the apical (intestinal lumen-facing side) to basolateral (blood-facing side). This effectively mimics the absorption of a drug across the intestinal wall.
- **Caco-2 Efflux Ratio:** Measures the rate of flux of a compound across polarized Caco-2 cell monolayers. The Efflux Ratio is determined by a ratio of the apparent permeability coefficient (Papp) in both directions. Ratios of ~1 indicate that a compound primarily traverses the cell membrane via passive (diffusional) transport. Ratios > 2 generally indicate *active* transport of the compound across the cellular membrane by membrane bound transporters (e.g. efflux by p-glycoprotein).
- **Mouse Plasma Protein Binding (MPPB):** Determines the concentration of free drug in plasma (as % Unbound). Drugs that are not bound to plasma can bind to target proteins and yield the desired therapeutic effect, making this parameter crucial to understanding drug distribution.

- **Mouse Brain Protein Binding (MBPB):** This measures the fraction of a drug not bound to proteins within brain tissue. The unbound fraction of a drug in the brain is considered pharmacologically active and able to interact with central nervous system (CNS) targets. MBPB helps assess CNS drug exposure and potential efficacy or side effects for neuroactive compounds. Reported as % Unbound.
- **Mouse Gastrocnemius Muscle Binding (MGMB):** This reflects the amount of drug free to act within **skeletal muscle tissue**, which is important for drugs targeting peripheral or muscular conditions (very important for the DM1 indication). Reported as % Unbound

Judging Criteria and Rules

We welcome submissions of any kind, including machine learning and physics-based approaches. You can employ pre-training approaches as you see fit. You are also free to reuse data from one portion of the challenge for others if it will assist you. You are free to incorporate data from external sources (*e.g*, public repositories) into your models and submissions.

You can submit to the HuggingFace portal multiple times (within a once-a-day rate limit, but up to the judges' final discretion), however, **only your latest submission will count** towards the intermediate and final leaderboards. In the spirit of our mission to promote open science, we encourage participants to share the code used to create their submissions. If not possible due to IP or other constraints, we require at a **minimum** that participants provide a short written report on the methodology used for submissions (based on the template [here](#)).

We encourage teams in industry to compete. We understand this isn't always possible in public-facing challenges, therefore, **we will allow teams to compete anonymously as a group using a pseudonym or alias.** A HuggingFace user account is required to compete and track multiple submissions. However, if you choose the anonymous participation option, the associated user account will not be visible to other participants. If choosing to compete anonymously, you can choose to reveal yourself post challenge if you wish.

The endpoints will be judged individually by *mean absolute error (MAE)*, while the overall leaderboard will be judged by the *macro-averaged relative absolute error (MA-RAE)*. For endpoints that are not already on a log scale (*e.g.* LogD), they will be transformed to a log scale to minimize the impact of outliers on evaluation. Relative absolute error (RAE) normalizes the MAE to the dynamic range of the test data, making the RAE comparable between endpoints, unlike MAE. We will estimate errors on the metrics using bootstrapping and use the statistical testing workflow outlined in [this paper](#) to determine if model performance is statistically distinct.

During the challenge, we will run a **live leaderboard** that will be updated after every submission. This live leaderboard will reflect the result of evaluating your submissions against a partial fraction of the test set (**validation split**). **A final leaderboard will be released after the submission window is closed**, showing the final submissions evaluated against the full test set. The competition is run with an open science ethos, and all queries and issues are encouraged to be discussed openly. The OMSF code of [conduct](#) applies to all communication around the challenge. The organizers reserve the right to exclude participants deemed to be acting in bad faith with respect to the integrity of the competition.

We plan to release a summary paper detailing the challenge results, to which all eligible teams will be invited to participate. Hope to see you there!

Timeline

The challenge timeline is as follows:

- September 12: First challenge announcement
- **October 10: Second announcement and sample data release**
- October 27: Challenge starts
- October-November: Online Q&A sessions and support via the Discord channel
- January 19, 2026: Submission closes and final leaderboard released
- January 26, 2026: Winners announced

Post-Challenge Virtual workshop

Participants with notable performance or learning outcomes will be invited to present their work at a special blind-challenge workshop, hosted by OpenADMET, to share their findings.

How to get started

Check out the teaser data and get excited! We will release a full tutorial prior to the start of the competition.

Questions or Ideas?

We'd love to hear from you! Whether you want to learn more, have ideas for future challenges, or wish to contribute data.

Join the **OpenADMET Discord** or contact us at **openadmet@omsf.io**.

Let's work together to transform ADMET modeling and accelerate drug discovery!

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